

CSCE 763 Final Exam Study Sheet

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1 Image Fundamentals (Ch. 2)

1.1 Pinhole / Geometric Imaging

Similar triangles relate object and image.

$$\frac{h_{\text{img}}}{h_{\text{obj}}} = \frac{f}{d} \Rightarrow h_{\text{img}} = h_{\text{obj}} \cdot \frac{f}{d}$$

where f = focal length (lens to sensor), d = object distance.

Worked example. Palm tree $h_{\text{obj}} = 15$ m at $d = 100$ m, $f = 17$ mm.

$$h_{\text{img}} = 15 \cdot \frac{0.017}{100} = 2.55 \times 10^{-3} \text{ m} = 2.55 \text{ mm}.$$

1.2 Pixel Relationships

- **4-adjacency** $N_4(p)$: up, down, left, right.
- **8-adjacency** $N_8(p)$: all 8 neighbors.
- **m-adjacency**: $q \in N_4(p)$ or $(q \in N_D(p) \text{ and } N_4(p) \cap N_4(q) \cap V = \emptyset)$. Removes path ambiguity of 8-adjacency.

1.3 Distance Measures

For $p = (x, y)$, $q = (s, t)$:

$$D_4 = |x - s| + |y - t| \quad (\text{city-block})$$

$$D_8 = \max(|x - s|, |y - t|) \quad (\text{chessboard})$$

$$D_E = \sqrt{(x - s)^2 + (y - t)^2}$$

1.4 Linear vs. Nonlinear

T is linear if $T(af + bg) = aT(f) + bT(g)$.

- **Linear**: average, sum, weighted sum, convolution, Laplacian, Sobel.
- **Nonlinear**: median, min, max, log, gamma, threshold, abs.

1.5 Image Arithmetic Applications

- **Addition / averaging K frames**: reduces *zero-mean* additive noise; $\sigma_{\bar{g}} = \sigma/\sqrt{K}$. Does not help DC-biased noise.
- **Subtraction**: change detection, mask-mode radiography, background removal.
- **Multiplication**: shading correction (divide by flat field), region-of-interest masking.

2 Intensity Transformations & Histograms (Ch. 3)

2.1 Point Transforms

Negative	$s = L - 1 - r$	invert; reveal detail in dark regions
Log	$s = c \log(1 + r)$	expand dark, compress bright (large dynamic range)
Inverse log	$s = c(e^r - 1)$	opposite of log
Gamma	$s = c r^\gamma$	$\gamma < 1$ brightens dark; $\gamma > 1$ darkens
nth power	$s = c r^n$	like gamma with $\gamma = n > 1$
nth root	$s = c r^{1/n}$	like gamma with $\gamma = 1/n < 1$

Choose c so output range is $[0, L - 1]$, e.g. $c = (L - 1) / \log L$ for log on $[0, L - 1]$.

2.2 Histogram Equalization (discrete)

Steps.

1. Count pixels per level: n_k for $k = 0, \dots, L - 1$.
2. Probability: $p_r(r_k) = n_k / (MN)$.
3. CDF mapping:

$$s_k = (L - 1) \sum_{j=0}^k p_r(r_j)$$

4. Round s_k to nearest integer in $[0, L - 1]$.
5. Replace each pixel of value r_k with s_k .

Tiny example. $L = 8$, $MN = 64$, $n = [8, 10, 10, 2, 12, 16, 4, 2]$. $p = [0.125, 0.156, 0.156, 0.031, 0.188, 0.250, 0.062, 0.031]$. CDF: $[0.125, 0.281, 0.438, 0.469, 0.656, 0.906, 0.969, 1.0]$. $s = 7 \cdot \text{CDF rounded} = [1, 2, 3, 3, 5, 6, 7, 7]$.

2.3 Histogram Matching

Goal: make f have the histogram of target z .

1. Equalize input: compute $T(r_k) = (L - 1) \sum p_r(r_j)$.
2. Equalize target: compute $G(z_q) = (L - 1) \sum p_z(z_j)$.
3. For each input level r_k , set $z_k = G^{-1}(T(r_k))$ — the smallest z_q with $G(z_q) \geq T(r_k)$.

3 Spatial Filtering (Ch. 3)

3.1 Convolution vs. Correlation

- **Correlation:** slide kernel over image, multiply-and-sum. *No flip*.
- **Convolution:** flip kernel by 180° , then correlate.
- Symmetric kernels (Gaussian, average, Laplacian) $\rightarrow$ convolution = correlation.

3.2 Convolution Properties

$f * g = g * f$ (commutative); $f * (g * h) = (f * g) * h$ (associative); $f * (g + h) = f * g + f * h$ (distributive).

3.3 Smoothing Kernels

Box / average 3×3 : $\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. **Weighted:** $\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ (Gaussian-ish).

3.4 Sharpening Kernels

Laplacian (4-conn): $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ **Laplacian** (8-conn): $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Sharpened image: $g(x, y) = f(x, y) - \nabla^2 f(x, y)$ (when center is negative) or $g = f + \nabla^2 f$ (center positive).

Unsharp mask: $g = f + k(f - \bar{f})$, with $k = 1$ standard, $k > 1$ *highboost*.

3.5 Gradient / First-Order Operators

Roberts (2×2): $G_x = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$, $G_y = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Prewitt (3×3): $G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$, $G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

Sobel (3×3 , weights center row/col more): $G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$, $G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$.

Magnitude: $|\nabla f| \approx |G_x| + |G_y|$ (or $\sqrt{G_x^2 + G_y^2}$). **Direction:** $\theta = \arctan(G_y/G_x)$.

3.6 Order-Statistic (nonlinear) Filters

- **Median:** salt & pepper.
- **Max:** removes pepper (dark).
- **Min:** removes salt (bright).
- **Midpoint** $\frac{1}{2}(\max + \min)$: Gaussian/uniform noise.
- **Alpha-trimmed mean:** mixed noise (drop $d/2$ from each end, average rest).

3.7 Filter Selection Cheat-Table

Symptom	Filter
Salt & pepper	Median (or alpha-trimmed)
Salt only	Min, harmonic, contraharmonic $Q < 0$
Pepper only	Max, contraharmonic $Q > 0$
Gaussian noise	Arithmetic mean, geometric mean, Wiener
Uniform noise	Arithmetic / midpoint
Sharpen edges	Laplacian, unsharp mask
Detect edges	Sobel, Prewitt, Canny

4 Fourier Transform (Ch. 4)

4.1 Continuous FT

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt, \quad f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

4.2 Sifting (impulse) Property

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

4.3 Properties

Property	Time	Frequency
Linearity	$af(t) + bg(t)$	$aF(\mu) + bG(\mu)$
Translation	$f(t - t_0)$	$e^{-j2\pi\mu t_0} F(\mu)$
Modulation	$e^{j2\pi\mu_0 t} f(t)$	$F(\mu - \mu_0)$
Scaling	$f(at)$	$\frac{1}{ a } F(\mu/a)$
Conjugation	$f^*(t)$	$F^*(-\mu)$
Symmetry	$F(t)$	$f(-\mu)$
Convolution	$f * g$	$F \cdot G$
Multiplication	$f \cdot g$	$F * G$

4.4 FT of a Rectangle (the trick for piecewise problems)

For $f(t) = A$ on $[a, b]$, 0 elsewhere:

$$F(\mu) = A \frac{e^{-j2\pi\mu a} - e^{-j2\pi\mu b}}{j2\pi\mu}$$

Equivalent (centered form), with width $W = b - a$, center $c = (a + b)/2$:

$$F(\mu) = AW \operatorname{sinc}(\pi\mu W) e^{-j2\pi\mu c}$$

where $\operatorname{sinc}(x) = \sin(x)/x$.

Worked piecewise example. $f(t) = \begin{cases} 2 & 3 < t < 7 \\ 4 & 7 < t < 9 \\ 0 & \text{else} \end{cases}$. Treat as two rectangles by linearity:

$$F(\mu) = 2 \cdot \frac{e^{-j6\pi\mu} - e^{-j14\pi\mu}}{j2\pi\mu} + 4 \cdot \frac{e^{-j14\pi\mu} - e^{-j18\pi\mu}}{j2\pi\mu}.$$

4.5 Convolution Theorem (key for filtering in frequency domain)

$$f(x, y) * h(x, y) \longleftrightarrow F(u, v) H(u, v)$$

Use: multiply spectra (cheap) instead of convolving (expensive). Apply lowpass / highpass / bandpass $H(u, v)$ in frequency, IFT back.

5 Image Restoration (Ch. 5)

5.1 Degradation Model

$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y) \xleftrightarrow{\text{FT}} G(u, v) = H(u, v)F(u, v) + N(u, v)$$

5.2 Noise PDFs

- Gaussian:** $p(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(z-\mu)^2/(2\sigma^2)}$.
- Salt-and-pepper (impulse):** $p(z) = P_a$ at $z = a$ (pepper, dark), P_b at $z = b$ (salt, bright), 0 else.
- Uniform:** $p(z) = 1/(b - a)$ on $[a, b]$.
- Rayleigh, Erlang/Gamma, Exponential:** also covered; rare on exam.

5.3 Mean-Filter Family

Filter	Formula (window S_{xy} , mn pixels)	Best for
Arithmetic	$\frac{1}{mn} \sum g(s, t)$	Gaussian, uniform; blurs
Geometric	$[\prod g(s, t)]^{1/mn}$	Gaussian; less blur
Harmonic	$\frac{\sum 1/g(s, t)}{\sum g^{Q+1}}$	salt + Gaussian; <i>fails on pepper</i>
Contraharmonic	$\frac{\sum g^{Q+1}}{\sum g^Q}$	$Q > 0$ pepper, $Q < 0$ salt

5.4 Restoration Filters: When to Use

Inverse: $\hat{F}(u, v) = G(u, v)/H(u, v)$.

- Needs H exactly, $H \neq 0$, no noise.
- Where $H \rightarrow 0$, noise blows up.

Wiener (min MSE):

$$\hat{F} = \left[\frac{1}{H} \frac{|H|^2}{|H|^2 + S_\eta/S_f} \right] G$$

- Needs $|H|^2$, power spectra S_η (noise) and S_f (signal).
- Approx: replace S_η/S_f with constant K .

Constrained Least Squares (CLS):

$$\hat{F} = \frac{H^*}{|H|^2 + \gamma|P|^2} G$$

- $P(u, v)$ = FT of Laplacian (smoothness constraint).
- Only needs noise mean and variance — no full spectra.
- Tune γ until $\|g - H\hat{f}\|^2 \approx \|\eta\|^2$.

6 Color Image Processing (Ch. 6)

6.1 Primaries

- **Light (additive):** R, G, B $\rightarrow$ mix $\rightarrow$ White. Secondary: Cyan, Magenta, Yellow.
- **Pigment (subtractive):** C, M, Y absorb light. Secondary: R, G, B.
- *Primary of one = secondary of the other.*

6.2 Chromaticity & Gamut

Chromaticity = hue + saturation (color quality, no brightness). **Gamut rule:** any color on the line segment between two reference colors in the chromaticity diagram is reproducible by a mixture of those two. Inside a triangle of three primaries $\Rightarrow$ producible by that system. Same color reachable by many RGB combos.

6.3 Color Models

- **RGB:** hardware, displays.
- **CMY/CMYK:** printing. $C = 1 - R$, $M = 1 - G$, $Y = 1 - B$. K extracts black: $K = \min(C, M, Y)$.
- **HSI:** matches human perception (Hue, Saturation, Intensity).

6.4 RGB to HSI

$$I = \frac{1}{3}(R + G + B), \quad S = 1 - \frac{3}{R + G + B} \min(R, G, B)$$

$$H = \begin{cases} \theta & B \leq G \\ 360^\circ - \theta & B > G \end{cases}, \quad \theta = \cos^{-1} \left[\frac{\frac{1}{2}[(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}} \right]$$

6.5 Which Color Model for Which Task

Task	Best model
Intensity modification	HSI (modify I, leave H,S)
Color complement	RGB ($s = L - 1 - r$ per channel)
Tonal correction	HSI
Color balancing	varies by image
Histogram processing	HSI (equalize I only)

6.6 Color Transformation General Form

$s_i = T_i(r_1, r_2, \dots, r_n)$ for $i = 1, \dots, n$.

7 Image Compression (Ch. 8)

7.1 Definitions

$$C = \frac{b}{b'}, \quad R = 1 - \frac{1}{C}$$

where b = bits original, b' = bits compressed.

Entropy: $H = -\sum_{k=0}^{L-1} p_r(r_k) \log_2 p_r(r_k)$.

Avg code length: $L_{\text{avg}} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$.

Coding efficiency: $\eta = H/L_{\text{avg}}$ (1 = optimal).

RMS error: $e_{\text{rms}} = \sqrt{\frac{1}{MN} \sum \sum [\hat{f} - f]^2}$.

SNR: $\text{SNR}_{\text{ms}} = \frac{\sum \sum \hat{f}^2}{\sum \sum [\hat{f} - f]^2}$.

7.2 Three Redundancies

- **Coding redundancy** — fixed-length code uses more bits than entropy. Fixed by Huffman / arithmetic coding.
- **Spatial / temporal redundancy** — neighboring pixels (or frames) correlated. Fixed by run-length, predictive coding, transform coding, motion comp.
- **Irrelevant information** — invisible to HVS. Fixed by quantization (lossy: JPEG, MP3).

7.3 Huffman Coding

Steps (encode).

1. List symbols with probabilities, sorted descending.
2. Take the two smallest, merge into a node with summed probability.
3. Reinsert sorted; repeat until one node remains (the root).
4. Label one branch 0 and the other 1 at every split.
5. Read each symbol's code from root $\rightarrow$ leaf.

Decode. Walk the tree from root using bits; emit symbol at leaf, restart.

Worked example. Symbols A,B,C,D,E with $p = 0.4, 0.2, 0.2, 0.1, 0.1$.

Merge D+E (0.2), then C+(DE) (0.4), then B+CDE (0.6), then A+rest. One valid set of codes: A=0, B=10, C=111, D=1100, E=1101. $L_{\text{avg}} = 2.2$, $H \approx 2.12$.

7.4 Arithmetic Coding

Encode.

1. Start with interval $[0, 1)$.
2. For each symbol, partition the current interval by symbol probabilities (in fixed order).
3. Narrow to the sub-interval of the actual symbol.
4. Repeat. Final code = any number in the final interval (the shortest binary fraction).

Mini example. Symbols a,b,c with $p = 0.5, 0.3, 0.2$. Sequence “ab”: start $[0, 1) \rightarrow$ after ‘a’ $[0, 0.5) \rightarrow$ partition again, after ‘b’ $[0.25, 0.4)$. Pick 0.3.

7.5 Run-Length Encoding (RLE)

Replace runs of the same value with (*value*, *count*) pairs.

Example. Bitstream 0000011111100 $\rightarrow (0, 5)(1, 6)(0, 2)$. Best when long runs (binary masks, simple cartoons). For natural images, combine with predictive coding first.

8 Edge-Based Segmentation (Ch. 10)

8.1 First- vs. Second-Order Edge Detection

- **First-order (gradient):** Roberts, Prewitt, Sobel. Detects edges as gradient maxima. Thicker edges; one operator per direction.
- **Second-order (Laplacian):** zero-crossings = edges. Double edge response. More noise sensitive.
- **Common issues:** both sensitive to noise; cannot handle scale changes (need LoG/DoG).

8.2 Laplacian of Gaussian (LoG) / Marr-Hildreth

1. Smooth with Gaussian: $G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/(2\sigma^2)}$.
2. Apply Laplacian: $\nabla^2[G_{\sigma} * f]$.
3. Find zero-crossings (edge locations).

Issues: “spaghetti” closed-loop zero-crossings; need multiple σ for scale.

8.3 Difference of Gaussians (DoG)

DoG = $G_{\sigma_1} - G_{\sigma_2}$, $\sigma_1 < \sigma_2$. Cheap LoG approximation; ratio $\sigma_2/\sigma_1 \approx 1.6$.

8.4 Canny Edge Detector

1. Smooth with Gaussian.
 2. Compute gradient magnitude and direction (Sobel).
 3. **Non-maximum suppression:** thin edges to 1 pixel by keeping local maxima along gradient direction.
 4. **Double threshold + hysteresis:** pixels above T_H are strong; pixels in $[T_L, T_H]$ are weak; keep weak only if connected to a strong.
- Optimal** (under Canny’s criteria): good detection, good localization, single response.

8.5 Hough Transform (line detection)

Line in polar form: $\rho = x \cos \theta + y \sin \theta$.

1. Build accumulator $A[\rho][\theta]$, all zero.
2. For each edge pixel (x, y) : for each θ in $[0, 180^\circ)$, compute ρ and increment $A[\rho][\theta]$.
3. Peaks in A correspond to lines. Each peak's (ρ, θ) defines one line.

Generalize: circle Hough uses (a, b, r) ; can detect any parameterized shape.

8.6 Thresholding

Choose T so $f(x, y) > T \rightarrow$ object, else background. **Otsu:** maximize between-class variance.

Factors that affect thresholding: (1) separation between histogram peaks, (2) noise level, (3) relative size of object/background, (4) illumination/reflectance uniformity.

9 Region-Based Segmentation (Ch. 10)

9.1 Region Growing

1. Pick seed pixel(s) (e.g., local maxima, user-defined).
2. Define predicate $Q(R)$ — e.g., $|f(p) - \mu_R| < T$ or intensity within ± 5 of seed.
3. Add neighbors of current region that satisfy Q and connectivity (4 or 8).
4. Repeat until no more pixels qualify; restart with new seed if needed.

9.2 Region Splitting and Merging (Quadtree)

1. Start with whole image as one region.
 2. **Split:** if $Q(R)$ false, split into 4 quadrants. Recurse.
 3. **Merge:** merge any pair of adjacent regions R_i, R_j if $Q(R_i \cup R_j)$ true.
 4. Stop when no further split or merge is possible.
- Q examples: $\sigma_R < T$, or $\max - \min < T$.

10 Mathematical Morphology (Ch. 9)

10.1 Sets and Operators

Reflection: $\hat{B} = \{w \mid w = -b, b \in B\}$.

Translation: $(B)_z = \{c \mid c = b + z, b \in B\}$.

Erosion: $A \ominus B = \{z \mid (B)_z \subseteq A\}$ — SE fits entirely inside A .

Dilation: $A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$ — reflected SE hits A .

Opening: $A \circ B = (A \ominus B) \oplus B$.

Closing: $A \bullet B = (A \oplus B) \ominus B$.

10.2 Effects (memorize the verbs)

Erosion — shrinks objects, removes small noise, breaks thin connections.

Dilation — grows objects, fills small holes, connects nearby pieces.

Opening — smooths outer contour, breaks narrow isthmuses, eliminates thin protrusions, removes small bright blobs.

Closing — fills narrow breaks/gaps, eliminates long thin gulfs, fills small dark holes.

10.3 Properties

- Dilation: commutative, associative, distributes over union.

- Erosion: distributes over intersection: $(A \cap B) \ominus C = (A \ominus C) \cap (B \ominus C)$.
- **Duality:** $A \oplus B = (A^c \ominus \hat{B})^c$.
- **Opening idempotent:** $(A \circ B) \circ B = A \circ B$. Opening shrinks: $A \circ B \subseteq A$.
- **Closing idempotent:** $(A \bullet B) \bullet B = A \bullet B$. Closing grows: $A \subseteq A \bullet B$.
- Monotone: $A \subseteq C \Rightarrow A \oplus B \subseteq C \oplus B$ and $A \ominus B \subseteq C \ominus B$.

10.4 How to Apply by Hand (binary image)

Erosion of A by B . For each pixel p in A , place SE origin at p . If *every* 1-pixel of B overlays a 1-pixel of A , output is 1; else 0.

Dilation of A by B . Reflect B to get $\hat{B}$. For each pixel p , place $\hat{B}$ origin at p . If *any* 1-pixel of $\hat{B}$ overlays a 1-pixel of A , output is 1; else 0.

Opening = erode then dilate (with the same SE).

Closing = dilate then erode.

10.5 Other Morphology

- **Hit-or-miss:** detect specific shape — erode by foreground SE and complement-erode by background SE; intersect.
- **Boundary:** $\beta(A) = A - (A \ominus B)$.
- **Hole filling:** iteratively dilate seed inside hole, intersect with A^c , until convergence.
- **Connected components:** same iterative dilation but seeded inside each component.
- **Convex hull, thinning, thickening, skeleton** — covered in slides; less likely on exam.

11 Last-Minute Quick-Recall List

- Pinhole: $h_i/h_o = f/d$
- $D_4 = |\Delta x| + |\Delta y|$, $D_8 = \max$, $D_E = \sqrt{\cdot}$
- Averaging K frames: $\sigma/\sqrt{K}$
- Negative: $L - 1 - r$; Log: $c \log(1 + r)$; Gamma: cr^γ
- HE: $s_k = (L - 1) \sum p_r(r_j)$
- Sobel: $\pm 1, \pm 2, \pm 1$
- Laplacian (4-conn): center -4
- Sharpen: $g = f - \nabla^2 f$ (neg center)
- FT: $\int f e^{-j2\pi\mu t} dt$
- Sifting: $\int f \delta(t - t_0) dt = f(t_0)$
- Conv $\leftrightarrow$ multiply
- $g = h * f + \eta$, $G = HF + N$
- Median = s&p; arithmetic = Gaussian
- HSI = perceptual; RGB = display
- $C = b/b'$, $R = 1 - 1/C$, $H = -\sum p \log_2 p$
- Huffman = sort+merge; RLE = (val,count)
- Canny: smooth, grad, NMS, hyst.
- Hough: $\rho = x \cos \theta + y \sin \theta$
- Open = erode-then-dilate
- Close = dilate-then-erode
- Duality: $A \oplus B = (A^c \ominus \hat{B})^c$